曲线积分与曲面积分

(A)

1．解：两点间直线段的方程为：$y=1-x$，$(0\le x\le 1)$

故 $ds=\sqrt{1+y'^2}dx=\sqrt{1+(-1)^2}dx=\sqrt{2}dx$

所以 $\int_L (x+y)dx=\int_0^1 (x+1-x)\sqrt{2}dx=\sqrt{2}$。

2．解：L 的参数方程为 $\begin{cases} x=\dfrac{1}{2}a\cos\theta+\dfrac{1}{2}a \\ y=\dfrac{1}{2}a\sin\theta \end{cases}$，$(0\le\theta\le 2\pi)$

则 $\sqrt{x^2+y^2}=\sqrt{\left(\dfrac{1}{2}a\cos\theta+\dfrac{1}{2}a\right)^2+\left(\dfrac{1}{2}a\sin\theta\right)}=\dfrac{1}{2}|a|\sqrt{2(1+\cos\varphi)}$

$=\dfrac{1}{2}|a|\sqrt{2\left(1+2\cos^2\dfrac{\theta}{2}-1\right)}=|a|\left|\cos\dfrac{\theta}{2}\right|$

$ds=\sqrt{x'^2+y'^2}d\theta=\sqrt{\left(\dfrac{-a}{2}\sin\theta\right)^2+\left(\dfrac{a}{2}\cos\theta\right)^2}=\dfrac{1}{2}|a|$

所以 $\int_L\sqrt{x^2+y^2}ds=\int_0^{2\pi}\dfrac{1}{2}a^2\left|\cos\dfrac{\theta}{2}\right|d\theta$

$=\dfrac{1}{2}a^2\left(\int_0^{\pi}\cos\dfrac{\theta}{2}d\theta-\int_{\pi}^{2\pi}\cos\dfrac{\theta}{2}d\theta\right)$

$=\dfrac{1}{2}a^2\left(2\sin\dfrac{\theta}{2}\Big|_0^{\pi}-2\sin\dfrac{\theta}{2}\Big|_{\pi}^{2\pi}\right)=2a^2$

3．解：$ds=\sqrt{x'^2+y'^2}dt=\sqrt{(at\cos t)^2+(at\sin t)^2}dt=atdt$

故 $\int_L(x^2+y^2)ds=\int_0^{2\pi}a^2\left[(\cos t+t\sin t)^2+(\sin t-t\cos t)^2\right]atdt$

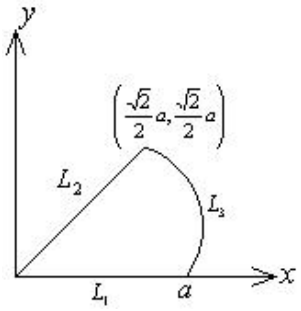
$=a^3\int_0^{2\pi}(t+t^3)dt=a^3\left(\dfrac{t^2}{2}+\dfrac{t^4}{4}\right)\Big|_0^{2\pi}=2\pi^2a^3(1+2\pi^2)$

4．解：如图 $\oint_L e^{\sqrt{x^2+y^2}}ds=\int_{L_1}e^{\sqrt{x^2+y^2}}ds+\int_{L_2}e^{\sqrt{x^2+y^2}}ds+\int_{L_3}e^{\sqrt{x^2+y^2}}ds$

L_1：$\begin{cases} x=x \\ y=0 \end{cases}$，$0\le x\le a$，$ds=\sqrt{1+0^2}dx=dx$

L_2：$\begin{cases} x=x \\ y=x \end{cases}$，$0\le x\le \dfrac{\sqrt{2}}{2}a$，$ds=\sqrt{1+1^2}dx=\sqrt{2}dx$

L_3：$\begin{cases} x=a\cos t \\ y=a\sin t \end{cases}$，$0\le x\le \dfrac{\pi}{4}$，



$$dx=\sqrt{x'^2+y'^2}dt=\sqrt{(-a\sin t)^2+(a\cos t)^2}dt=adt$$

$$\therefore \oint_L e^{\sqrt{x^2+y^2}}ds=\int_0^a e^x dx+\int_0^{\frac{\sqrt{2}}{2}a}e^{\sqrt{2}x}\sqrt{2}dx+\int_0^{\frac{\pi}{4}}e^a adt$$

$$=e^x\,|_0^a+e^{\sqrt{2}x}\Big|_0^{\frac{\sqrt{2}}{2}a}+ae^a\frac{\pi}{4}=e^a\left(2+\frac{\pi}{4}a\right)-2$$

5．解：$x^{\frac{4}{3}}+y^{\frac{4}{3}}=a^{\frac{4}{3}}\left(\cos^4 t+\sin^4 t\right)$

$$ds=\sqrt{x'^2+y'^2}dt=\sqrt{\left(-3a\cos^2 t\sin t\right)^2+\left(3a\sin^2 t+\cos t\right)^2}$$

$$=\sqrt{9a^2\sin^2 t\cos^2 t}dt=3a\sin t\cos t dt$$

$$\therefore \int_L\left(x^{\frac{4}{3}}+y^{\frac{4}{3}}\right)ds=3a^{\frac{7}{3}}\int_0^{\frac{\pi}{2}}\left(\cos^4 t+\sin^4 t\right)\sin t\cos t dt$$

$$=3a^{\frac{7}{3}}\left(-\frac{1}{6}\cos 6t+\frac{1}{6}\sin^6 t\right)\Big|_0^{\frac{\pi}{2}}=4a^{\frac{7}{3}}$$

6．解：$ds=\sqrt{x'^2+y'^2+z'^2}dt=\sqrt{(-a\sin t)^2+(a\cos t)^2+a^2}dt=\sqrt{2}adt$

$$\therefore \int_L\frac{z^2}{x^2+y^2}ds=\int_0^{2\pi}\frac{a^2t^2}{(a\cos t)^2+(a\sin t)^2}\sqrt{2}adt=2\sqrt{a}\int_0^{2\pi}t^2dt$$

$$=\sqrt{2}a\frac{1}{3}t^3\,|_0^{2\pi}=\frac{8}{3}\sqrt{2}a\pi^3\text{。}$$

7．解：$\int_L xydx=\int_{-1}^1 y^2y\left(y^2\right)'dy=2\int_{-1}^1 y^4dy=2\dfrac{1}{5}y^5\Big|_{-1}^1=\dfrac{4}{5}$

8．解：直线段 AB 的方程为 $\dfrac{x}{3}=\dfrac{y}{2}=\dfrac{z}{1}$，化成参数方程为

$x=3t$，$y=2t$，$z=t$，t 从 1 变到 0

故 $\int_L x^3dx+3xy^2dy-x^2ydz$

$$=\int_1^0\left[(3t)^3\cdot 3+3t(2t)^2\cdot 2-(3t)^2\cdot 2t\right]dt$$

$$=87\int_1^0 t^3dt=-\frac{87}{4}$$

9．解：直线的参数方程为

$x=1+t$，$y=1+2t$，$z=1+3t$ $(0\le t\le 1)$

$$\int_L xdx+ydy+(x+y-1)dz$$

$$=\int_0^1[(1+t)+2(1+2t)+3(1+t+1+2t-1)]dt$$

$$=\int_0^1(6+14t)dt=13$$

10．解：$\int_L(2a-y)dx-(9-y)dy$

$$=\int_0^{2\pi}\{[2a-a(1-\cos t)]a(1-\cos t)-[a-a(1-\cos t)]a\sin t\}dt$$

$$=a^2\int_0^{2\pi}(1-\cos^2 t-\cos t\sin t)dt=a^2\int_0^{2\pi}\left[\frac{1}{2}(1-\cos 2t)-\frac{1}{2}\sin 2t\right]dt$$

$$=a^2\int_0^{2\pi}\frac{1}{2}dt=a^2\pi$$

11．解：1)原式$=\int_1^2\left[(y^2+y)2y+(y-y^2)\right]dy$

$$=\int_1^2(2y^3+y^2+y)dy=\left(\frac{1}{2}y^4+\frac{1}{3}y^3+\frac{1}{2}yh2\right)_1^2=\frac{34}{3}$$

2)原式$=\int_0^1\{[(2t^2+t+1)+(t^2+1)](4t+1)+[(t^2+1)-(2t^2+t+1)]2t\}dt$

$$=\int_0^1(10t^3+5t^2+9t+2)dt$$

$$=\left(\frac{10}{4}t^4+\frac{5}{3}t^2+\frac{9}{2}t+2\right)dt=\left(\frac{10}{4}t^4+\frac{5}{3}t^2+\frac{9}{2}t^2+2t\right)\Big|_0^1=\frac{131}{12}$$

12．解：1）L 的方向余弦 $\cos\alpha\frac{3}{5}$，$\cos\beta\frac{4}{5}$

$$\int_L P(x,y)dx+Q(x,y)dy=\int_L\left[\frac{3}{5}P(x,y)+\frac{4}{5}Q(x,y)\right]ds$$

2）$ds=\sqrt{1+(2x)^2}\,dx$，$\cos\alpha=\dfrac{dx}{dx}=\dfrac{1}{\sqrt{1+4x^2}}$

$$\cos\beta=\sin\alpha=\sqrt{1-\frac{1}{1+4x^2}}=\frac{2x}{\sqrt{1+4x^2}}$$

故$\int_L P(x,y)dx+Q(x,y)dy=\int\dfrac{P(x,y)+2xQ(x,y)}{\sqrt{1+4x^2}}ds$

3）$ds=\sqrt{1+\dfrac{(1-x)^2}{2x-x^2}}dx$，$\cos\alpha=\dfrac{dx}{ds}=\sqrt{2x-x^2}$

$$\cos\beta=\sin\alpha=\sqrt{1-2x+x^2}=1-x$$

故$\int_L P(x,y)dx+Q(x,y)dy=\int_L\left[\sqrt{2x-x^2}\,P(x,y)+(1-x)Q(x,y)\right]ds$

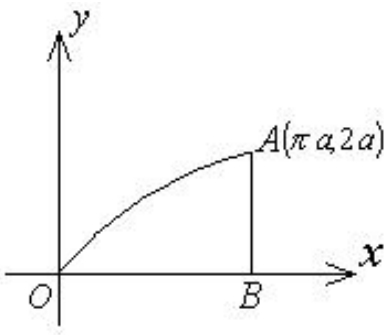
13．解：因为$\dfrac{\partial P}{\partial y}=\dfrac{\partial Q}{\partial x}=e^x\cos y-m$

故原积分与路径无关，于是

原式$=\int_{\overline{OB}}+\int_{\overline{BA}}$

$=\int_0^{\pi a}0dx+\int_0^{2a}\left(e^{\pi a}\cos y-m\pi a\right)dy$

$=e^{\pi a}\sin 2a-2\pi ma^2$。



14．解：$P=x^4+4xy^\lambda$，$Q=6x^{\lambda-1}y^2-5y^4$，由$\dfrac{\partial P}{\partial y}=\dfrac{\partial Q}{\partial x}$，得

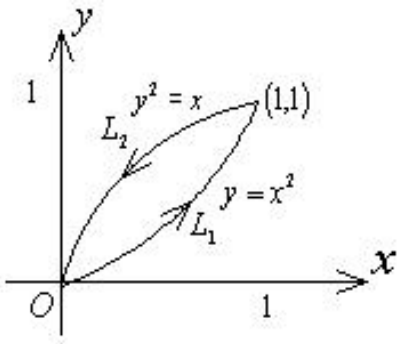
$4\lambda xy^{\lambda-1}=6(\lambda-1)x^{\lambda-2}y^2$，解得$\lambda=3$

故当$\lambda=3$时，所给积分与路径无关

$$\int_{(0,0)}^{(1,2)}\left(x^4+4xy^3\right)dx+\left(6x^2y^2-5y^4\right)dy$$

$$=\int_0^1\left(x^4+4x\cdot 0\right)dx+\int_0^2\left(6\cdot 1\cdot y^2-5y^4\right)dy=\frac{79}{5}$$

取 $\overline{AC}+\overline{CB}$ 计算，其中 $A(0,0)$，$C(1,0)$，$B(1,2)$



15．解：原式 $\int_{L_1}+\int_{L_2}$

$$=\int_0^1\left[\left(2x^3-x^2\right)+\left(x+x^4\right)2x\right]dx$$

$$+\int_1^0\left[\left(2y^3-y^4\right)2y+\left(y^2+y^2\right)dy\right]$$

$$=\int_0^1\left(2x^5+2x^3+x^2\right)dx$$

$$-\int_0^1\left(-2y^5+4y^4+2y^2\right)dy=\frac{1}{30}$$

又 $\iint_D\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dxdy=\iint_D(1-2x)dx=\int_0^1dy\int_{y^2}^{\sqrt{y}}(1-2x)dx=\frac{1}{30}$

$\therefore\ \iint_D\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dxdy=\oint_L Pdx+Qdy$

16．解取 $P=-y$，$Q=x$，$\frac{\partial P}{\partial y}=-1$，$\frac{\partial Q}{\partial x}=1$ 可得面积

$$A_1=\iint_D dxdy=\frac{1}{2}\oint_L xdy-ydx$$

设 A_1 为在第 I 象限部分的面积，由图形的对称性所求面积

$$A=4A_1=4\frac{1}{2}\oint xdy-ydx$$

$$=2\int_0^{\frac{\pi}{2}}\left[a\cos^3t\cdot3a\sin^2t\cos t-a\sin^2t\cdot3a\cos^2t(-\sin t)\right]dt$$

$$=ba^2\int_0^{\frac{\pi}{2}}\sin^2t\cos^2tdt=\frac{3}{8}\pi a^2$$

注：还可利用 $\iint_D dxdy=\oint_L xdy=\oint_L ydx$

17．解：$P=6xy^2-y^3$，$Q=6x^2y=3xy^2$

$$\frac{\partial P}{\partial y}=12xy-3y^2,\quad\frac{\partial Q}{\partial x}=12xy-3y^2$$

因为$\frac{\partial Q}{\partial x}=\frac{\partial P}{\partial y}$，所以积分与路径无关

取路径$(1,2)\to(3,2)\to(3,4)$

原式$=\int_1^3(24x-8)dx+\int_2^4(54y-9y^2)dy=236$

18．解：$\frac{\partial Q}{\partial x}=2x\sin x+x^2\cos y-2ye^x$，$\frac{\partial P}{\partial y}=x^2\cos x+2x\sin x-2ye^x$

原式$=\iint_D\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dxdy=0$。

19．解：$\frac{\partial Q}{\partial x}=3$，$\frac{\partial P}{\partial y}=-1$

原式$=\iint_D\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dxdy=\iint_D[3-(-1)]dxdy=\iint_D 4dxdy$

$=\int_0^3 dx\int_0^{\frac{2}{3}x}4dy=\int_0^3\frac{8}{3}xdx=12$

20．解：1）$\frac{\partial Q}{\partial x}=2x=\frac{\partial P}{\partial y}$，故$2xydx+x^2dy$是某个$u(x,y)$的全微分。

$u(x,y)=\int_{(0,0)}^{(x,4)}2xydx+x^2dy=\int_0^x 0dx+\int_0^y x^2dy-x^2y$

2　　）　　$\frac{\partial Q}{\partial x}=3x^2+16x=\frac{\partial P}{\partial y}$　　，

$u(x,y)=\int_{(0,0)}^{(x,y)}(3x^2y+8xy^2)dx+(x^3+8x^2y+12ye^y)dy$

$=\int_0^x 0dx+\int_0^y(x^3+8x^2y+12ye^y)dy$

$=x^3y+4x^2y^2+12(ye^y-e^y)+12$

21．解：D_{xy}：$x^2+y^2\le 2$，$dx=\sqrt{1+z_x^2+z_y^2}dxdy=\sqrt{1+4x^2+4y^2}dxdy$

故原式$=\iint_{D_{xy}}(x^2+y^2)\sqrt{1+4x^2+4y^2}dxdy$

$=\int_0^{2\pi}d\theta\int_0^{\sqrt{2}}[(r\cos\theta)^2 9+(r\sin\theta)^2]\sqrt{1+4(r\cos\theta)^2+4(r\sin\theta)^2}rdr$

$$=\int_0^{2\pi}d\theta\int_0^{\sqrt{2}}r^2\sqrt{1+4r^2}rdr=2\pi\frac{1}{2}\int_0^{\sqrt{2}}r^2\sqrt{1+4r^2}d(rh2)$$

$$\overset{r^2=u}{=\!=}\pi\int_0^{\sqrt{2}}u\sqrt{1+4u}du=\frac{149}{30}\pi$$

22．解：原式$=\iint\limits_{D_{xy}}|x||y|\left(x^2+y^2\right)\sqrt{1+z_x'+z_y'}dxdy$

$$=4\iint\limits_{D_{xyI}}xy\left(x^2+y^2\right)\sqrt{1+4\left(x^2+y^2\right)}dxdy$$

这里D_{xyI}为D_{xy}在第一象限部分

$$=4\int_0^{\frac{\pi}{2}}d\theta\int_0^1r^4\sin\theta\cos\theta\sqrt{1+4r^2}rdr=4\int_0^{\frac{\pi}{2}}\frac{1}{2}\sin2\theta d\theta\int_0^1 7^4\sqrt{1+4r^2}rdr$$

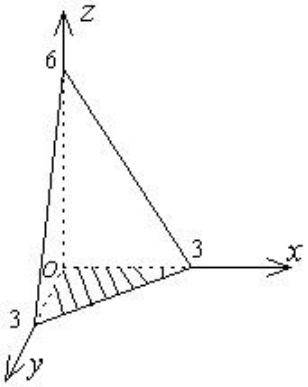
$$=\int_0^1r^4\sqrt{1+4r^2}d\left(r^2\right)\overset{\sqrt{1+4r^2}=t}{=\!=}\frac{1}{32}\int_1^{\sqrt{5}}\left(t^4-2t^2+1\right)t^2dt=\frac{125\sqrt{5}-1}{420}$$

23．解：$z=6-2x-2y$，$ds=\sqrt{1+(-2)^2}dxdy=3dxdy$

原式$=\iint\limits_{D_{xy}}\left(2xy-2x^2-x+6-2x-2y\right)3dxdy$

$$=3\int_0^3dx\int_0^{3-x}\left(6-3x-2x^2+2xy-2y\right)dy$$

$$=-\frac{27}{4}$$



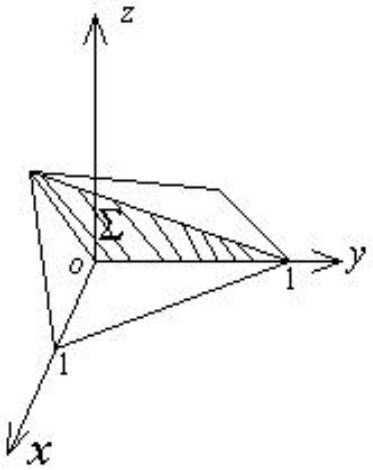
24．解：$M=\iint\limits_{\Sigma}zds=\iint\limits_{D_{xy}}\frac{1}{2}\left(x^2+y^2\right)\sqrt{1+x^2+y^2}dxdy$

$$=\int_0^{2\pi}d\theta\int_0^{\sqrt{2}}\frac{1}{2}r^2\sqrt{1+r^2}7dr=\frac{2\pi}{15}\left(6\sqrt{3}+1\right)$$

25．解：平面$z=x$这部分的面积

$$S=\iint\limits_D\sqrt{1+z_x'^2+z_y'^2}dxdy=\iint\limits_D\sqrt{2}dxdy$$

$$=\sqrt{2}\int_0^1 dx\int_0^{1-x} dy=\frac{\sqrt{2}}{2}$$



因而 $\bar{x}=\frac{1}{S}\iint_\Sigma x ds=\frac{2}{\sqrt{2}}\int_0^1 x dx\int_0^{1-x}\sqrt{2}dy\frac{1}{3}$

$$\bar{y}=\frac{1}{S}\iint_\Sigma y ds=\frac{2}{\sqrt{2}}\int_0^1 dx\int_0^{1-x} y\sqrt{2}dy\frac{1}{3}$$

$$\bar{z}=\frac{1}{S}\iint_\Sigma z ds=\frac{2}{\sqrt{2}}\iint_\Sigma x ds\frac{1}{3}$$

故重心坐标为 $\left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right)$

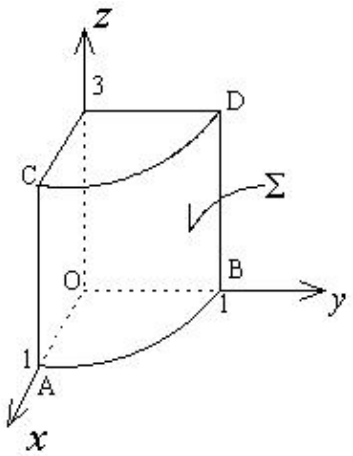
26. 解：因为曲面积分Σ有向曲面，所以 $\iint_\Sigma R(x,y,z)dxdy=\pm\iint_{D_{xy}} R(x,y,0)dxdy$

当积分曲面取在Σ的上侧时为正号，取在下侧时为负号

27，解：$D_{xy}=\overset{\frown}{AB}$，面积为 0，$\iint_\Sigma z dxdy=0$

$D_{yz}=\{(0,y,z)\,|\,x=0,0\le y\le 1,0\le z\le 3\}$，

$D_{zx}=\{(x,0,z)\,|\,0\le x\le 1,y=0,0\le z\le 3\}$



原式 $=\iint_{D_{yz}}\sqrt{1-y^2}dydz+\iint_{D_{zx}}\sqrt{1-x^2}dzdx$

$$=\int_0^3 dz\int_0^1\sqrt{1-y^2}dy+\int_0^3 dz\int_0^1\sqrt{1-x^2}dx$$

$$=2\cdot\left[\frac{y}{2}\sqrt{1-yh2}+\frac{1}{2}\arcsin y\right]_0^1=\frac{3}{2}\pi\ 。$$

28. 解：根据轮换对称，只要计算 $\iint_\Sigma z^2 dxdy$

D_{xy}：$(x-a)^2+(y-b)^2\le R^2$

注意到：$z-e=\pm\sqrt{R^2-(x-a)^2+(y-b)^2}$，再利用极坐标可得

$$-\iint_\Sigma z^2 dxdy=\iint_{D_{xy}}\left[c+\sqrt{R^2-(x-a)^2-(y-b)^2}\right]dxdy-$$

$$-\iint_{D_{xy}}\left[c-\sqrt{R^2-(x-a)^2-(y-b)^2}\right]dxdy$$

$$=4e\iint_{D_{xy}}\sqrt{R^2-(x-a)^2-(y-b)^2}\,dxdy$$

$$=4e\int_0^{2\pi}d\theta\int_0^R\sqrt{R^2-r^2}\,rdr$$

$$=8\pi e\left[-\frac{1}{3}\left(R^2-r^2\right)^{\frac{3}{2}}\right]_0^R=\frac{8}{3}\pi R^3e$$

于是原式$=\frac{8}{3}\pi R^3(a+b+c)$

29．解：原式$=\iint_\Sigma(P\cos\alpha+Q\cos\beta+R\cos\gamma)ds$，这里$\cos\alpha$，$\cos\beta$，$\cos\gamma$是$\Sigma$的法向理$\vec{n}$的方向余弦而$\Sigma$是平面$3x+2y+2\sqrt{3}z=6$在第一卦限部分的上侧$\cos\gamma>0$，取$\vec{n}=\{3,2,2\sqrt{3}\}$。

$$\cos\alpha=\frac{3}{\sqrt{3^2+2^2+\left(2\sqrt{3}\right)^2}}=\frac{3}{5}，\quad\cos\beta=\frac{2}{5}，\quad\cos\gamma=\frac{2\sqrt{3}}{5}$$

故原式$=\iint_\Sigma\left(\frac{3}{5}R+\frac{2}{5}Q+\frac{2\sqrt{3}}{5}R\right)dx$。

30．解：1）$\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}=2x+2y+2z$

$$原式=\iiint_\Omega\left(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}\right)dxdydz=2\iiint_\Omega(x+y+z)dxdydz$$

$$=2\int_0^a dx\int_0^a dy\int_0^a(x+y+z)dz=2\int_0^a dx\int_0^a\left(ax+ay+\frac{a^2}{2}\right)dy$$

$$=2\int_0^a\left(a^2x+\frac{a^3}{2}+\frac{a^3}{2}\right)dx=3a^4$$

20 $P=(y-z)x$，$Q=0$，$R=x-y$

$$\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}=y-z$$

故原式 $=\iiint_{\Omega}(y,z)dxdydz=\int_0^{2\pi}d\theta\int_0^1 rdr\int_0^3(x\sin\theta-z)dz=-\frac{9}{2}\pi$ 。

31．解：$\Phi=\oiint_{\Sigma}Pdydz+Qdzdx+Rdxdy$

$$=\iiint_{\Omega}\left(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}\right)dxdydz$$

$$=\iiint_{\Omega}(2+x^2-2xz)dxdydz=\int_0^a dx\int_0^a dy\int_0^a(2+x^2-2xz)dz$$

$$=a\int_0^a(2a+ax^2-a^2x)dx-a^3\left(2-\frac{a^2}{6}\right)$$

2）$\Phi=\oiint_S\vec{\alpha}d\vec{s}=\oiint_{\Sigma}(x-y+z)dydz+(y-z+x)dzdx+(z-x+y)dxdy$

$=\iiint_{\Omega}(1+1+1)dxdydz=3\cdot\frac{4}{3}\alpha bc=4\pi abc$ 。

32．解：$P=e^{xy}$ ，$Q=\cos(xy)$，$R=\cos(xz^2)$

$\frac{\partial P}{\partial x}=ye^{xy}$，$\frac{\partial Q}{\partial y}=-x\sin(xy)$，$\frac{\partial R}{\partial z}=-2xz\sin(xz^2)$

故 $div\vec{\alpha}=\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}=ye^{xy}-x\sin(xy)-2xz\sin(xz^2)$

33．解：取Σ为平面 $x+y+z=0$，被所围成的部分的上侧，Σ 的面积为 πa^2，Σ 的单位法向量为

$$\vec{n}=\{\cos\alpha,\cos\beta,\cos\gamma\}=\left\{\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right\}$$

$$\text{原式}=\iint_{\Sigma}\begin{vmatrix}\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}}\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ x & y & z\end{vmatrix}ds=\iint_{\Sigma}\left(-\frac{1}{\sqrt{3}}-\frac{1}{\sqrt{3}}-\frac{1}{\sqrt{3}}\right)ds$$

$$=-\frac{3}{\sqrt{3}}\iint_{\Sigma}ds=-\sqrt{3}\pi a^2\ 。$$

34．证：平面 $y=z$ 的单位法向理 $\vec{n}=\{\cos\alpha,\cos\beta,\cos\gamma\}=\left\{0,\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}\right\}$

由斯托克斯公式得

$$左边=\iint_{\Sigma}\begin{vmatrix}\cos\alpha & \cos\beta & \cos\gamma\\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z}\\ y^2 & xy & xz\end{vmatrix}ds=\frac{1}{\sqrt{2}}\iint_{\Sigma}(y-z)ds$$

$$=\frac{1}{\sqrt{2}}\iint_{D_{xy}}(y-z)dxdy=0$$

35．解：闭曲线Γ是xoy平面上的圆周$x^2+y^2=4$(逆时针方向)，它的参数方程为$x=2\cos\theta$，$y=2\sin\theta$，$z=0$ $(0\le\theta\le 2\pi)$，故环流量为

$$\oint_{\Gamma}Rdx+Qdy+Rdz=\oint_{\Gamma}(x-z)dx+(x^3+yz)dy-3xy^2dz$$

$$=\int_0^{2\pi}\left[2\cos\theta(-2\sin\theta)+8\cos^3\theta 2\cos\theta\right]d\theta$$

$$=-4\int_0^{2\pi}\sin\theta\cos\theta d\theta+16\int_0^{2\pi}\cos^4\theta d\theta=0+12\pi=12\pi.$$

36．解：$rot\vec{\alpha}=\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z}\\ z+\sin y & -(z-x\cos y) & 0\end{vmatrix}=\vec{i}+\vec{j}$。

37．解：证平面$x+y+z=\frac{3}{2}a$合科立方体内的部分为Σ，它在oxy平面上的射影为D_{xy}，面积为$\frac{3}{4}a^2$，取平面的上侧，单位法向量$\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$，于是由斯托克斯公式得

$$原式=\iint_{\Sigma}\begin{vmatrix}\dfrac{1}{\sqrt{3}} & \dfrac{1}{\sqrt{3}} & \dfrac{1}{\sqrt{3}}\\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z}\\ y^2-z^2 & z^2-x^2 & x^2-y^2\end{vmatrix}ds=-4\iint_{\Sigma}(x+y+z)\frac{1}{\sqrt{3}}dx$$

$$=-6a\iint_{\Sigma}\frac{1}{\sqrt{3}}ds=-6a\iint_{D_{xy}}dxdy=-6a\frac{3}{4}a^2=-\frac{9}{2}a^3。$$

(B)

1．解：L 的参数方程 $\begin{cases} x=\dfrac{1}{2p}t^2 \\ y=t \end{cases}$，则

$$ds=\sqrt{x'^2+y'^2}\,dt=\sqrt{\left(\frac{1}{p}t\right)^2+1^2}\,dt=\frac{1}{p}\sqrt{t^2+p^2}\,dt$$

所以 $\int_L y\,ds=\int_0^{y_0} t\frac{1}{p}\sqrt{t^2+p^2}\,dt=\frac{1}{p}\frac{1}{3}\left(t^2+p^2\right)^{\frac{3}{2}}\Big|_0^{y_0}=\frac{1}{3p}\left[\left(y_0^2+p^2\right)^{\frac{3}{2}}-p^3\right]$

2．解：$ds=\sqrt{x'^2+y'^2}\,dt=\sqrt{a^2(1-\cos t)^2+a^2\sin^2 t}\,dt$

$$=a\sqrt{2(1-\cos t)}=1\sqrt{2\left[1-\left(1-2\sin^2\frac{t}{2}\right)\right]}=2a\left|\sin\frac{t}{2}\right|$$

所以 $\int_L y^2\,ds=\int_0^{2\pi} a^2(1-\cos t)^2\,2a\left|\sin\frac{t}{2}\right|dt$

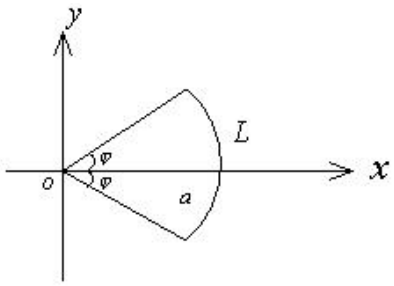
$$=8a^3\int_0^{2\pi}\sin^4\frac{t}{2}\left|\sin\frac{t}{2}\right|dt=8a^3\int_0^{2\pi}\left(1-\cos^2\frac{t}{2}\right)\sin\frac{t}{2}\,dt$$

$$=-16a^3\left(\cos\frac{t}{2}-\frac{2}{3}\cos^3\frac{t}{2}+\frac{1}{5}\cos^5\frac{t}{2}\right)\Big|_0^{2\pi}=\frac{256}{15}a^3$$

3．解：取坐标系如图，设重心坐标为 $G(\bar{x},\bar{y})$，由扇形的对称性可知 $\bar{y}=0$，

又 $\bar{x}=\dfrac{\int_L \rho x\,ds}{\int_L \rho\,ds}=\dfrac{1}{24a}\int_{-4}^{4} a\cos\theta\, a\,d\theta$

$$=\frac{a}{24}\sin\theta\,|_{-4}^{4}=\frac{a\sin\varphi}{\varphi}$$



4．解：

$$ds=\sqrt{x'^2+y'^2+z'^2}=\sqrt{(\cos t-t\sin t)+(\sin t+t\cos t^2)+1}\,dt$$

$$=\sqrt{2+t^2}\,dt$$

所以 $\int_L z\,ds=\int_0^{t_0} t\sqrt{2+t^2}\,dt=\frac{1}{3}\left(2+t^2\right)^{\frac{3}{2}}\Big|_0^{t_0}=\frac{1}{3}\left[\left(2+t_0^2\right)^{\frac{3}{2}}-2\sqrt{2}\right]$

5．解 $x^2+y^2+z^2-\left(e^t\cos t\right)^2+\left(e^t\sin t\right)^2+\left(e^t\right)^2=2e^{2t}$

$$ds=\sqrt{x'^2+y'^2+z'^2}\,dt$$

$$=\sqrt{\left(e^t\cos t-e^t\sin t\right)^2+\sqrt{\left(e^t\sin t+e^t\cos t\right)^2+\left(e^t\right)^2}}\,dt$$

$$=\sqrt{3e^{2t}}\,dt=\sqrt{3}e^t$$

所以$\int_L\frac{1}{\sqrt{x^2+y^2+z^2}}ds=\int_0^2\frac{1}{2e^{2t}}\sqrt{3}e^t$

$$=\frac{\sqrt{3}}{2}\int_0^2e^{-t}dt=-\frac{\sqrt{3}}{2}e^{-t}\Big|_0^2=\frac{\sqrt{3}}{2}\left(1-e^{-2}\right)$$

6．解：$M=\int_L\rho(x,y,z)ds$

$$=\int_0^{2\pi}\left(a^2\cos^2t+a^2\sin^2t+k^2t^2\right)\cdot\sqrt{a^2\sin^2t+a^2\cos^2t+k^2}\,dt$$

1）$I_z=\int_L\left(x^2+y^2\right)\rho(x,y,z)ds=\int_0^{2\pi}\left(x^2+y^2\right)\left(x^2+y^2+z^2\right)ds$

$$=\int_0^{2\pi}a^2\left(a^2+k^2t^2\right)\sqrt{a^2+k^2}\,dt=\frac{2}{3}\pi a^2\sqrt{a^2+k^2}\left(3a^2+4\pi^2k^2\right)$$

2）$\bar{x}=\frac{1}{M}\int_Lx\rho(x,y,z)ds=\frac{1}{M}\int_0^{2\pi}a\cos t\left(a^2+k^2t^2\right)\sqrt{a^2+k^2}\,dt$

$$=\frac{6ak^2}{3a^2+4\pi^2k^2}$$

$$\bar{y}=\frac{1}{M}\int_Ly\rho(x,y,z)ds=\frac{1}{M}\int_0^{2\pi}a\sin t\left(a^2+k^2t^2\right)\sqrt{a^2+k^2}\,dt$$

$$=\frac{-6\pi ak^2}{3a^2+4\pi^2k^2}$$

$$\bar{z}=\frac{1}{M}\int_Lz\rho(x,y,z)ds=\frac{1}{M}\int_0^{2\pi}kt\left(a^2+k^2t^2\right)\sqrt{a^2+k^2}\,dt$$

$$=\frac{3\pi k\left(a^2+2\pi^2khh2\right)}{3a^2+4\pi^2k^2}$$

7．解：由$x=t$，$y=t^2$，$z=t^3$得

$dx=dt$，$dy=2tdt=2xdt$，$dz=3t^2dt=3ydt$

$ds-\sqrt{1+4x^2+9y^2}\,dt$

故 $\cos\alpha=\dfrac{dx}{ds}=\dfrac{1}{\sqrt{1+4x^2+9y^2}}$

$\cos\beta=\dfrac{dy}{ds}=\dfrac{2x}{\sqrt{1+4x^2+9y^2}}$

$\cos\gamma=\dfrac{dz}{ds}=\dfrac{3y}{\sqrt{1+4x^2+9y^2}}$

故 $\int_L Pdx+Qdy+Rdz=\int_L\dfrac{P+2xQ+3yR}{\sqrt{1+4x^2+9y^2}}ds$

8．解：圆周的参数方程为 $x=a\cos t$，$y=a\sin t\ (0\le t\le 2\pi)$

故 $\oint_L\dfrac{(x+y)dx-(x-y)dy}{x^2+y^2}$

$=\dfrac{1}{a^2}\int_0^{2\pi}[(a\cos t+a\sin t)(-a\sin t)-(a\cos t-a\sin t)(a\cos t)]dt$

$=\dfrac{1}{a^2}\int_0^{2\pi}-a^2dt=-2\pi$

9．解：$\int_L ydx+zdy+xdz=\int_0^{2\pi}[a\sin t(-a\sin t)+bt(a\cos t)+(a\cos t)b]dt$

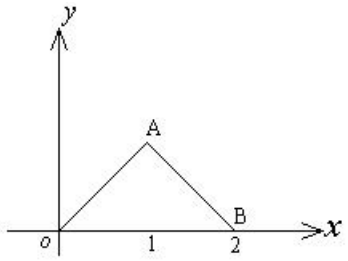
$=\int_0^{2\pi}(-a^2\sin^2 t+abt\cos t+ab\cos t)dt=-\pi a^2$

10．解：如图 $C=\overline{OA}+\overline{OB}$，

OA：$y=x$，　AB：$y=z-x$

故原式 $=\int_0^1(x^2+x^2)dx+\int_1^2[x^2+(2-x)^2]dx$

$+\int_1^2[x^2-(2-x)^2]d(2-x)=\dfrac{4}{3}$



11．解：由于 $P=2xe^y+y$，　$Q=x^2y+x-2y$

又 $\dfrac{\partial P}{\partial y}=\dfrac{\partial Q}{\partial x}=2xe^y+1$，故曲线积分与路径无关，取折线 $(1,0)\to(2,0)\to(2,1)$，

则原式 $=\int_1^2 2xdx+\int_0^1(4e^y+2-2y)dy=4e$ 。

12．解：由于 $P=\dfrac{x}{\left(x^2+y^2\right)^{3/2}}$，$Q=\dfrac{y}{\left(x^2+y^2\right)^{3/2}}$，

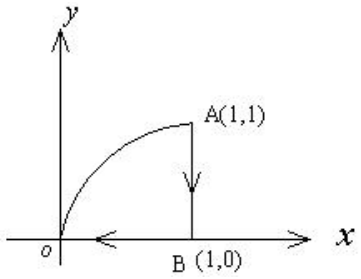
又 $\dfrac{\partial P}{\partial y}=\dfrac{\partial Q}{\partial x}=-3xy\left(x^2+yh2\right)^{-5/2}$ 故当路径不过原点时，该曲线积分与路径无关，取折线 $(1,1)\to(2,1)\to(2,2)$，得

原式 $\int_1^2\dfrac{xdx}{\left(x^2+1\right)^{3/2}}+\int_1^2\dfrac{ydy}{\left(4+y^2\right)^{3/2}}=\dfrac{\sqrt{2}}{4}$

13．解：取参数方程 $x=a\cos t$，$y=\sin t\ (0\le t\le 2\pi)$

面积 $A=\dfrac{1}{2}\oint_L xdy-ydx=\dfrac{1}{2}\int_0^{2\pi}ab\left(\cos^2 t+\sin^2 t\right)dt=\pi ab$

14．解：L 不是闭曲线，要用格林公式，先得补添路径，使其封闭，如图



$\int_L+\int_{AB}+\int_{BO}=-\iint_D\left(\dfrac{\partial P}{\partial x}-\dfrac{\partial P}{\partial y}\right)dxdy$，

因为 $\dfrac{\partial P}{\partial x}-\dfrac{\partial P}{\partial y}=0-1-(-1)=0$

故 $\int_L+\int_{AB}+\int_{BO}=0$，所以

原式 $\int_L+\int_{AB}+\int_{BO}=\int_0^1-\left(1+\sin^2 y\right)dy+\int_0^1x^2dx=-\dfrac{7}{6}+\dfrac{1}{4}\sin 2$

15．解：作代换 $y=tx$，得曲线的参数方程

$x=\dfrac{3at}{1+t^3}$，$y=\dfrac{3at^2}{1+t^3}$，由于 $dx=\dfrac{3a\left(1-2t^3\right)}{\left(1+t^3\right)^2}dt$，$dy=\dfrac{3at\left(2-t^3\right)}{\left(1+t^3\right)^2}dt$

从而 $xdy-ydx=\dfrac{9a^2t^2}{\left(1+t^3\right)^2}dt$，故面积

$S=\dfrac{1}{2}\oint_L xdy-ydx=\dfrac{9a^2}{2}\int_0^{+\infty}\dfrac{t^2}{\left(1+t^3\right)^2}dt=\dfrac{3a^2}{2}\left(-\dfrac{1}{1+t^3}\right)\Big|_0^{+\infty}=\dfrac{3}{2}a^2$

16．解：由于 $x=y=0$ 时，被积函数无意义，故 L 所包围的区域不满足格林公式的条件，作一小圆挖去原点 $(0,0)$，作逆时针方向的圆周 l：

$x=r\cos\theta$，　$y=r\sin\theta$，　$0\le\theta\le 2\pi$

使l全部补L所包围，在L和l为边界的区域D内，根据格要公式，有

$$\iint_{D_1}\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dxdy=\oint_L\frac{ydx-xdy}{2\left(x^2+y^2\right)}-\oint_l\frac{ydx-xdy}{2\left(x^2+y^2\right)}$$

$\because$　$\dfrac{\partial P}{\partial y}=\dfrac{x^2-y^2}{\left(x^2+y^2\right)^2}=\dfrac{\partial Q}{\partial x}$，故上式为零

$$\therefore\oint_L\frac{ydx-xdy}{2\left(x^2+y^2\right)}=\oint_l\frac{ydx-xdy}{2\left(x^2+yh2\right)}=\int_0^{2\pi}\frac{-r^2\sin^2\theta-r^2\cos^2\theta}{2r^2}d\theta$$

$$=-\frac{1}{2}\int_0^{2\pi}d\theta=-\pi\ 。$$

17．解：D_{xy}：$x^2+y^2\le 2$，　$ds=\sqrt{1+z_x^2+z_y^2}dxdy=\sqrt{1+4x^2+4y^2}dxdy$

$$原式=\iint_{D_{xy}}3\left[2-\left(x^2+y^2\right)\right]\sqrt{1+4x^2+4yh2}dxdy$$

$$=3\int_0^{2\pi}d\theta\int_0^{\sqrt{2}}\left(2-r^2\right)\sqrt{1+4r^2}rdr=\frac{111}{16}\pi$$

18．解：D_{xy}：$x^2+y^2\le 2ax$，　$z=\sqrt{x^2+y^2}$

$$ds=\sqrt{1+z_x^2+z_y^2}dxdy=\sqrt{1+\frac{x^2}{x^2+y^2}+\frac{y^2}{x^2+y^2}}dxdy=\sqrt{2}dxdy$$

$$原式=\sqrt{2}\iint_{D_{xy}}\left[xy+(x+y)\sqrt{x^2+y^2}\right]dxdy$$

$$=\sqrt{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}d\theta\int_0^{2a\cos\theta}\left[r^2\sin\theta\cos\theta+r^2(\sin\theta+\cos\theta)\right]rd\theta$$

$$=\sqrt{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(\sin\theta\cos\theta+\sin\theta+\cos\theta)\frac{1}{4}\left(2a\cos^4\theta\right)d\theta=\frac{64}{15}\sqrt{2}a^4\ 。$$

19．解：半球壳Σ的方程为$z=\sqrt{a^2-x^2-y^2}$

D_{xy}：　$x^2+y^2\le ah2$

$$ds=\sqrt{1+z_x^2+z_y^2}ds=\frac{a}{\sqrt{a^2-x^2-y^2}}dxdy$$

$$I_z=\iint_{\Sigma}\left(x^2+y^2\right)\rho_0 ds=\rho_0\iint_{D_{xy}}\frac{a\left(x^2+y^2\right)}{\sqrt{a^2-x^2-y^2}}dxdy$$

$$=a\rho_0\int_0^{2\pi}d\theta\int_0^a\frac{r^2}{\sqrt{a^2-r^2}}rdr=\frac{4}{3}\rho_0\pi a^4\ 。$$

20．解：质量为 $M=\iint_{\Sigma}\rho_0 ds=\sqrt{2}\rho_0\iint_{x^2+y^2\le ax}dxdy=\sqrt{2}\rho_0\left(\frac{a}{2}\right)^2=\frac{\sqrt{2}\pi a^2\rho_0}{4}$

从而垂心的坐标为

$$\bar{x}=\frac{1}{M}\sqrt{2}\rho_0\iint_{x^2+y^2\le ax}xdxdy=\frac{4}{\pi a^2}\int_0^a xdx\int_{\sqrt{ax-x^2}}^{\sqrt{ax-x^2}}dy$$

$$=\frac{8}{\pi a^2}\int_0^a x\sqrt{ax-x^2}dx=\frac{8}{\pi a^2}\int_{-\frac{a}{2}}^{\frac{a}{2}}\left(\frac{a}{2}+t\right)\sqrt{\left(\frac{a}{2}\right)^2-x^2}dt$$

$$=\frac{8}{\pi a}\int_0^{\frac{\pi}{2}}\sqrt{\left(\frac{a}{2}\right)^2-t^2}dt=\frac{a}{2}$$

$$\bar{y}=\frac{1}{M}\sqrt{2}\rho_0\int_0^a dx\int_{-\sqrt{ax-x^2}}^{\sqrt{ax-x^2}}ydy=0$$

$$\bar{z}=\frac{1}{M}\sqrt{2}\rho_0\iint_{x^2+y^2\le ax}zdxdy=\frac{4}{\pi a^2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}d\theta\int_0^{a\cos\theta}r^2dr=\frac{8a}{3\pi}\int_0^{\frac{\pi}{2}}\cos^3\theta d\theta=\frac{16a}{9\pi}$$

即重心坐标为 $\left(\frac{a}{2},0,\frac{16a}{9\pi}\right)$。

21．解：由于曲面 $z=\sqrt{x^2+y^2}$ 得 $x^2+y^2+z^2=a^2$ 分成上下两部分，记成 $S_{上}$，$S_{下}$，又由 $\begin{cases}x^2+y^2+z^2=a^2\\ z=\sqrt{x^2+yh2}\end{cases}$

解得：$z=\frac{a}{\sqrt{2}}$，$\sqrt{x^2+y^2}=\frac{a}{\sqrt{2}}$，所以

$$I=\iint_{S_{上}}\left(x^2+y^2\right)ds+\iint_{S_{下}}0ds$$

$$=\iint_{D_{xy}}\left(x^2+y^2\right)\frac{a}{\sqrt{a^2-x^2-y^2}}dxdy$$

$$=\int_0^{2\pi}d\theta\int_0^{\frac{a}{\sqrt{2}}}\frac{ar^3}{\sqrt{a^2-r^2}}dr=2\pi a^4\int_0^{\frac{\pi}{4}}\sin^2\theta d\theta=\frac{\pi a^4}{6}\left(8-5\sqrt{2}\right)$$

22．解：证 z 在 xoy， yoz， zox 平面上的部分分别为 Σ_1， Σ_2， Σ_3，在 $x+y+z=1$ 面上的部分为 Σ_4。

$$\iint_{\Sigma_1}xzdxdy+xydydz+yzdzdx=\iint_{\Sigma_1}xzdxdy=-\iint_{D_{xy}}x\cdot 0dxdy=0$$

$$\iint_{\Sigma_2}xzdxdy+xydydz+yzdzdx=\iint_{\Sigma_2}xydydz=-\iint_{D_{yz}}0\cdot ydydz=0$$

$$\iint_{\Sigma_3}xzdxdy+xydydz+yzdzdx=\iint_{\Sigma_2}yzdzdx=-\iint_{D_{zx}}0\cdot zdzdx=0$$

故原式 $=\iint_{\Sigma_4}xzdxdy+xydydz+yzdzdx$

$$=3\iint_{\Sigma_4}xzdxdy=3\iint_{D_{xy}}x(1-x-y)dxdy=3\int_0^1dx\int_0^{1-x}(1-x-y)dy=\frac{1}{8}$$

(另解：可求得 $\iint_{\Sigma}xzdxdy=\frac{1}{24}$，由对称性可得原式 $=\frac{1}{8}$ 也可用高斯公式)

23．解：D_{xy}：$\frac{x^2}{a^2}+\frac{y^2}{b^2}\le 1$，$z=\pm c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}$ 由轮换对称，只要计算积分 $\iint_{\Sigma}\frac{1}{z}dxdy$ 再利用广义极坐标可得

$$\iint_{\Sigma}\frac{1}{2}dxdy=\iint_{D_{xy}}\frac{1}{c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}}dxdy-\iint_{D_{xy}}\frac{-1}{c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}}dxdy$$

$$=\frac{2}{c}\iint_{D_{xy}}\frac{1}{\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}}dxdy=\frac{2ab}{c}\int_0^{2\pi}d\theta\int_0^1\frac{r}{\sqrt{1-r^2}}dr$$

$$=\frac{4\pi ab}{c}\left(-\sqrt{1-r^2}\right)\Big|_0^1=4\pi\frac{ab}{c}$$

于是原式 $=4\pi\left(\frac{bc}{a}+\frac{ac}{b}+\frac{ab}{c}\right)=\frac{4\pi}{abc}\left(b^2c^2+a^2c^2+a^2b^2\right)$。

24．解：证Σ_1，Σ_2分别为锥面的底面和侧面而$\cos\alpha$，$\cos\beta$，$\cos\gamma$为锥面外法线的方向余弦D_{xy}：$x^2+y^2\le h^2$，则

$$\iint_{\Sigma_1}(y-z)dydz+(z-x)dxdy+(x-y)dxdy$$

$$=\iint_{D_{xy}}(x-y)dxdy=\int_0^{2\pi}d\varphi\int_0^h r^2(\cos\theta-\sin\theta)dr=0$$

又对Σ_2上的任一点(x,y,z)有

$$\frac{\cos\alpha}{x}=\frac{\cos\beta}{y}=\frac{\cos\gamma}{-r}$$

故ds在各坐标平面上射影分别为

$\cos\gamma ds=-dxdy$，$\cos\alpha dx=-\frac{x}{z}\cos\gamma ds=\frac{x}{2}dxdy$，$\cos\beta ds=-\frac{y}{z}\cos\gamma ds=\frac{y}{z}dxdy$

于是$\iint_{\Sigma_2}(y-z)dydz+(z-xdxdy)+(x-y)dxdy$

$$\iint_{S_2}[(y-z)\cos\alpha+(z-x)\cos\beta+(x-y)\cos\gamma]ds$$

$$=\iint_{D_{xy}}\left[\frac{x}{2}(y-z)+\frac{y}{z}(z-x)-(x-y)\right]dxdy$$

$$=-2\iint_{D_{xy}}(x-y)dxdy=0$$

故原式$=0+0=0$

25．证：由格林第一公式得

$$\iiint_\Omega u\Delta v dxdydz=\oiint_\Sigma u\frac{\partial v}{\partial n}dx-\iiint_\Omega\left(\frac{\partial u}{\partial x}\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\frac{\partial v}{\partial y}+\frac{\partial u}{\partial z}\frac{\partial v}{\partial z}\right)dxdydz$$

同理

$$\iiint_\Omega v\Delta u dxdydz=\oiint_\Sigma u\frac{\partial u}{\partial n}dx-\iiint_\Omega\left(\frac{\partial u}{\partial x}\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\frac{\partial v}{\partial y}+\frac{\partial u}{\partial z}\frac{\partial v}{\partial z}\right)dxdydz$$

两式相减得：

$$\iiint_{\Omega}(u\Delta v - v\Delta u)dxdydz = \oiint_{\Sigma}\left(u\frac{\partial v}{\partial n} - v\frac{\partial v}{\partial n}\right)ds \text{ 。}$$

26．解：设 $c = \Gamma + \overline{BA}$，其中 $\overline{BA}$ 为从 B 到 A 的直线段，则 c 为封闭曲线，由斯托克斯公式得

$$\oint_c Pdx + Qdy + Rdz = \iint_{\Sigma}\begin{vmatrix} dydz & dzdx & dxdy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 - yz & y^2 - xz & z^2 - xy \end{vmatrix}$$，其中 Σ 是以 c 为边界且与 c 构成右手系的任曲面。

$$\therefore \quad \int_L = \oint_C - \int_{\overline{BA}} = \oint_c + \int_{\overline{AB}} = \int_{\overline{AB}}$$

$$= \int_0^h (z^2 - ak0)dz = \frac{h^3}{3}$$

27．证：$gradu = \frac{\partial u}{\partial x}\vec{i} + \frac{\partial u}{\partial y}\vec{j} + \frac{\partial u}{\partial z}\vec{k}$

$$rot(gradu) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \end{vmatrix} = \left(\frac{\partial^2 u}{\partial z \partial y} - \frac{\partial^2 u}{\partial y \partial z}\right)\vec{i} - \left(\frac{\partial^2 u}{\partial z \partial x} - \frac{\partial^2 u}{\partial x \partial z}\right)\vec{j}$$

$$+\left(\frac{\partial^2 u}{\partial y \partial x} - \frac{\partial^2 u}{\partial x \partial y}\right)\vec{k} = 0$$

(C)

1．解：$ds = \sqrt{1 + \frac{a^2}{a^2 - xh2} + \frac{a^4}{4(a^2 - x^2)^2}}dx = \frac{3a^2 - 2x^2}{2(a^2 - x^2)}dx$，$|x_0| < 0$

于是当 $x_0 \geq 0$ 时，有

$$S = \int_0^{x_0} \frac{3a^2 - x^2}{2(a^2 - x^2)}dx = \frac{a}{4}\ln\frac{a + x_0}{a - x_0} + x_0 = |z_0| + |x_0|$$

当 $x_0 < 0$ 时，有

$$S = \int_{x_0}^0 \frac{3a^2 - 2x^2}{2(a^2 - x^2)}dx = \frac{a}{4}\ln\frac{a + x_0}{a - x_0} - x_0 = |z_0| + |x_0|$$

故当$|x_0|<a$时，有$S=|Z_0|+|x_0|$

2．解：$ds=\sqrt{1+\sin^2\frac{x}{a}}dx=ch\frac{x}{a}dx$，于是

$$\int_L\frac{1}{y^2}ds=\int_{-\infty}^{+\infty}\frac{ch\frac{x}{a}}{a^2ch^2\frac{x}{a}}dx=\frac{1}{a}\int_{-\infty}^{+\infty}\frac{d\left(sh\frac{x}{a}\right)}{1+sh^2\frac{x}{a}}$$

$$=\frac{1}{a}\arctan\left(sh\frac{x}{a}\right)\Big|_{-\infty}^{+\infty}=\frac{\pi}{a}$$

3．解：$ds=\sqrt{e^{2t}(\cos t-\sin t)^2+e^{2t}(\sin t+\cos t)^2+e^{2t}}\,dt=\sqrt{3}e^t dt$

质量为$M=\int_{-\infty}^{0}\sqrt{3}e^t dt=\sqrt{3}$

于是垂心坐标为

$$x_0=\frac{1}{m}\int_{-\infty}^{+\infty}e^t\cos t\sqrt{3}e^t dt=\int_{-\infty}^{0}e^{2t}\cos t dt=\frac{2\cos t+\sin t}{5}e^{2t}\Big|_{-\infty}^{0}=\frac{2}{5}$$

$$y_0=\frac{1}{m}\int_{-\infty}^{0}e^t\cos t\sqrt{3}e^t dt=\int_{-\infty}^{0}e^{2t}\sin t dt=\frac{2\sin t-\cos t}{5}e^{2t}\Big|_{-\infty}^{0}=-\frac{1}{5}$$

$$z_0=\frac{1}{m}\int_{-\infty}^{0}e^t e^t\sqrt{3}e^t dt=\int_{-\infty}^{0}e^{2t}dt=\frac{1}{2}$$

4．解：$\because\ \oint_{C+\overline{BA}}=\int_C+\int_{\overline{BA}}$，　$\therefore\ \int_C=\oint_{C+BA}+\int_{\overline{AB}}$

但$\int_{\overline{AB}}=\int_{-R}^{R}0dx=0$，又

$$\oint_{C+\overline{BA}}Pdx+Qdy=\iint_D\left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right)dxdy=\iint_D\left(4+\frac{2y}{\sqrt{R^2+x^2}}-\frac{2y}{\sqrt{R^2+x^2}}\right)dxdy$$

$$=4\iint_D dxdy=4\frac{\pi}{2}R^2=2\pi R^2$$

$\therefore$原式$=2\pi R^2$

5．解：$P=\frac{1+y^2f(xy)}{y}$，　$Q=\frac{x}{y^2}\left[y^2f(xy)-1\right]$

$$\frac{\partial P}{\partial y}=\frac{y\left[2yf(xy)+y^2f'(xy)x\right]-1-y^2f(xy)}{y^2}=\frac{y^2f(xy)+xy^3f'(xy)-1}{y^2}$$

$$\frac{\partial Q}{\partial x}=\frac{1}{y^2}\left\{\left[y^2f(xy)-1\right]+xy^2f'(xy)y\right\}=\frac{y^2f(xy)+xy^3f'(xy)-1}{y^2}$$

故当 $y\neq 0$ 时，$\frac{\partial P}{\partial y}=\frac{\partial Q}{\partial x}$，因此只要路径不过 x 轴，点 A 到点 B 的曲线积分与路径无关，取路径 $A\left(3,\frac{2}{3}\right)\to C\left(1,\frac{2}{3}\right)\to B(1,2)$，有

$$\text{原式}=\int_3^1\frac{1+\frac{4}{9}f\left(\frac{2}{3}x\right)}{\frac{2}{3}}dx+\int_{\frac{2}{3}}^2\frac{1}{y^2}\left[y^2f(y)-1\right]dy$$

$$=\frac{3}{2}\int_3^1dx+\frac{2}{3}\int_3^1f\left(\frac{2}{3}x\right)dx+\int_{\frac{2}{3}}^2f(y)dy-\int_{\frac{2}{3}}^2\frac{1}{y^2}dy$$

$$=\frac{3}{2}(1-3)+\int_2^{\frac{2}{3}}f(y)dy+\int_{\frac{2}{3}}^2f(y)dy+\left.\frac{1}{y}\right|_{\frac{2}{3}}^2=-4$$

6．解：$x\neq 0$ 时，有 $P=1-\frac{y^2}{x^2}\cos\frac{y}{x}$，$Q=\sin\frac{y}{x}+\frac{y}{x}\cos\frac{y}{x}$

$$\frac{\partial P}{\partial y}=-\frac{2y}{x^2}\cos\frac{y}{x}+\frac{y^2}{x^2}\sin\frac{y}{x}$$

$$\frac{\partial Q}{\partial x}=-\frac{y}{x^2}\cos\frac{y}{x}-\frac{y^2}{x^2}\cos\frac{y}{x}\neq\frac{y}{x^3}\sin\frac{y}{x}-\frac{2y}{x^2}\cos\frac{y}{x}+\frac{y^2}{x^3}\sin\frac{y}{x}$$

改右半平面 $\Omega=\{(x,y)\setminus x>0\}$，由于 Ω 是单连通区域，且在其上 $\frac{\partial Q}{\partial x}=\frac{\partial P}{\partial y}$，故在 Ω 上的是某函数 $u(x,y)$ 的全微分，且可取

$$u(x,y)=\int_1^x\left(1-\frac{y^2}{x^2}\cos\frac{y}{x}\right)dx+\int_\pi^y(\sin y+y\cos y)dy$$

$$=\left(x+y\sin\frac{y}{x}\right)\Bigg|_1^x+y\sin y\,|_\pi^y=x-1+y\sin\frac{y}{x}$$

于是原式$=\left(x-1+y\sin\frac{y}{x}\right)\Big|_{(1,\pi)}^{(2,\pi)}=\pi+1$

7．解：$P=x+xy\sin x$，$Q=\frac{f(x)}{2}$

$$\frac{\partial P}{\partial y}=x\sin x,\quad \frac{\partial Q}{\partial x}=\frac{xf'(x)-f(x)}{xh2}$$

即$f'(x)-\frac{1}{x}f(x)=x^2\sin x$

解此一阶线性微分方程得

$$f(x)=e^{\int\frac{1}{x}dx}\left[\int x^2\sin xe^{-\int\frac{1}{x}dx}dx+C\right]$$

$$=x(\sin x-x\cos xe)$$

由$f\left(\frac{\pi}{2}\right)=0$得$e=-1$，故所求函数为

$$f(x)=x(\sin x-x\cos x-1)$$

8．解：所求的功$W=\int_{(1,1)}^{(2,4)}\frac{xdx+ydy}{(x^2+y^2)^{3/2}}$，$P=\frac{x}{(x^2+y^2)^{3/2}}$，$Q=\frac{y}{(x^2+y^2)^{3/2}}$

$$\frac{\partial P}{\partial y}=-\frac{3}{2}\frac{2xy}{(x^2+y^2)^{3/2}},\quad \frac{\partial Q}{\partial y}=-\frac{3}{2}\frac{\partial xy}{(x^2+y^2)^{3/2}}$$

当$x^2+y^2\neq 0$时，此积分与路径无关

故$W=\int_1^2\frac{x}{(x^2+y^2)^{3/2}}dx+\int_1^4\frac{y}{(4+y^2)^{2/3}}dy$

$$=\frac{1}{\sqrt{1+x^2}}\Big|_1^2-\frac{1}{\sqrt{4+y^2}}\Big|_1^4=\frac{1}{\sqrt{2}}-\frac{1}{2\sqrt{5}}$$

9．解：由格林公式各

$$I=\iint_D(3-3x^2-3y^2)dxdy=\int_0^{2\pi}d\theta\int_0^R(3-3x^2)xdx$$

$$=6\pi\left(\frac{R^2}{2}-\frac{R^4}{4}\right)=3\pi R^2\left(1-\frac{R^2}{2}\right)$$

1）当 $R=0$（舍去），$R=\sqrt{2}$ 时，$I=0$

2）由 $I_R=6\pi R-6\pi R^3=0$，得 $R=0$（舍去），$r=1$

$I_R''=6\pi-18\pi R^2$，$I''|_{R=1}=-12\pi<0$

故当 $R=1$ 时，I 取最大值，$I(1)=\frac{3}{2}\pi$

10．解：补上 Σ_1：$x^2+y^2\le 1$，$z=1$，上侧由高斯公式

$$I=\iiint_{\Omega}3dxdydz-\iint_{\Sigma_1}x\left(1+x^2z\right)dydz+y\left(1-x^2z\right)dzdx+z\left(1-x^2z\right)dxdy$$

$$=3\cdot\frac{1}{3}\pi\cdot1^2\cdot1-\iint_{x^2+y^2\le1}\left(1-x^2\right)dxdy$$

$$=\pi-\pi+\int_0^{2\pi}d\theta\int_0^1 xx^2\cos^2\theta d\theta=\frac{\pi}{4}$$

11．解：由对称性可知

原式 $=8\iint_{\Sigma_1}xyzds$，Σ_1：$x+y+z=1$

而 $\iint_{\Sigma_1}xyzds=\iint_D xy(1-xy)\sqrt{1+z_x^2+z_y^2}dxdy$

$$=\sqrt{3}\int_0^1 dx\int_0^{1-x}xy(1-x-y)dy=\sqrt{3}\int_0^1\left[(1-x)x\frac{y^2}{2}-x\frac{y^3}{3}\right]_0^{1-x}dx$$

$$=\sqrt{3}\int_0^1\left[\frac{x}{2}(1-x)^3-\frac{x}{3}(1-x)^3\right]dx=\sqrt{3}\int_0^1\frac{x}{6}(1-x)^3dx=\frac{\sqrt{3}}{120}$$

故原式 $=8\frac{\sqrt{3}}{120}=\frac{\sqrt{3}}{15}$

12．解：取 Σ_1：$x=1$，$y^2+z^2=1$ 方向与 x 轴同上，则

$$I=\oiint_{\Sigma+\Sigma_1}2(1+x)dydz-\iint_{\Sigma_1}2(1+x)dydz=2\iiint_{\Omega}dxdydz-\iint_{\Sigma_1}4dydz$$

$$=2\int_0^{2\pi}d\theta\int_0^1 rdr\int_{r^2}^1 dx-4\pi=4\pi\int_0^1\left(r-r^3\right)dr-\pi=-3\pi$$

13．解：利用格林公式

原式

$$=\oint_{L+\overline{OA}}(y+2xy)dx+\left(x^2+2x+y\right)dy-\int_{\overline{OA}}(y+2xy)dx+\left(x^2+2x+y^2\right)dy$$

$$=\iint_D[(2x+2)-(1-2x)dxdy]-0=\iint_D dxdy=4\pi$$

14．解：$P=\dfrac{3y-x}{(x+y)^3}$，$\dfrac{\partial P}{\partial y}=\dfrac{6(x-y)}{(x+y)^4}$，$Q=\dfrac{y-3x}{(x+y)^3}$，$\dfrac{\partial Q}{\partial y}=\dfrac{6(x-y)}{(x+y)^4}$

当$x+y\neq 0$时，有$\dfrac{\partial P}{\partial y}=\dfrac{\partial Q}{\partial x}$，积分与路径无关

$$\int_{(1,0)}^{(2,3)}\frac{(3y-x)dx+(y-3x)dy}{(x+y)^3}=\int_1^2\frac{-x}{x^3}dx+\int_0^3\frac{y-6}{(2+y)^3}dy$$

$$=\frac{1}{x}\Big|_1^2+\int_0^3\frac{1}{(y+2)^2}dy-8\int_0^3\frac{1}{(y+2)^3}dy$$

$$=-\frac{1}{2}+\left(-\frac{1}{5}+\frac{1}{2}\right)+4\left(\frac{1}{25}-\frac{1}{4}\right)=\frac{26}{25}$$

15．解：$I_z=\iint_S\left(x^2+y^2\right)ds=\iint_{D_{xy}}\left(x^2+y^2\right)\sqrt{2}dxdy$

$$=\int_0^{2\pi}d\theta\int_0^h\sqrt{2}r^3dr=\frac{\sqrt{2}}{2}\pi h^4$$

16．解：$P(x,y)=\dfrac{y^2+2xy+ax^2}{\left(x^2+y^2\right)^2}$，$Q(x,y)=-\dfrac{x^2+2xy+by^2}{\left(x^2+y^2\right)^2}$

$$\frac{\partial P}{\partial y}=\frac{2y+2x}{\left(x^2+y^2\right)^2}-\frac{4y\left(y^2+2xy+ax^2\right)}{\left(x^2+y^2\right)^3},$$

$$\frac{\partial Q}{\partial x}=\frac{-(2y+2x)}{\left(x^2+y^2\right)^2}+\frac{4y\left(x^2+2xy+by^2\right)}{\left(x^2+y^2\right)^3}$$

令$\dfrac{\partial P}{\partial y}=\dfrac{\partial Q}{\partial x}$，得

$$x^3+x^2y+xy^2+y^3\equiv x^3+y^3+(+b)xy^2+(2+a)x^2y$$

比较系数得，$a=-1$，$b=-1$

$$\therefore du=\frac{y^2+2xy-x^2}{\left(x^2-y^2\right)^2}dx-\frac{x^2+2xy-y^2}{\left(x^2+y^2\right)^2}dy$$

$$\therefore u(x,y)=\int_{(1,1)}^{(x,y)}\frac{y^2+2xy-x^2}{x^2+y^21^2}dx-\frac{x^2+2xy-y^2}{\left(x^2+y^2\right)^2}dy$$

$$=\int_1^x\frac{1-2x-x^2}{\left(1+x^2\right)^2}dx-\int_1^y\frac{x^2+2xy-y^2}{\left(x^2+y^2\right)^2}dy$$

$$=\int_1^x\frac{d\left(1+x^2\right)}{\left(1+x^2\right)^2}+2\int_1^x\frac{dx}{\left(1+x^2\right)^2}-\int_1^x\frac{dx}{1+x^2}$$

$$+\int_1^y\frac{dy}{x^2+y^2}-2x\int_1^y\frac{(x+y)dy}{\left(x^2+y^2\right)^2}$$

$$=\frac{x-1}{x^2+1}+\frac{1}{1+x^2}-\frac{x}{1+xh2}-\frac{y}{x^2+y^2}+\frac{x}{x^2+y^2}=\frac{x-y}{x^2+y^2}$$

故$u(x,y)$的形式$\frac{x-y}{x^2+y^2}+l$

17. 解：$\oiint_\Sigma=\iint_{\Sigma_1}+\iint_{\Sigma_2}+\iint_{\Sigma_3}$，其中$\Sigma_1$，$\Sigma_2$，$\Sigma_3$分别是$\Sigma$在$z=1$，$z=2$，$z=x^2+y^2$上的曲面块。

$$\iint_{\Sigma_1}=-\iint_{D_1}\frac{e}{\sqrt{x^2+y^2}}dxdy=\lim_{\varepsilon\to0^+}\int_0^{2\pi}d\theta\int_\varepsilon^1 edr=-2\pi e$$

$$\iint_{\Sigma_2}=-\iint_{D_2}\frac{e^{\sqrt{2}}}{\sqrt{x^2+y^2}}dxdy=\lim_{\varepsilon\to0^+}\int_0^{2\pi}d\theta\int_\varepsilon^{\sqrt{2}}e^{\sqrt{2}}dr=2\sqrt{2}\pi e^{\sqrt{2}}$$

$$\iint_{\Sigma_3}=-\iint_{D_3}\frac{e^{\sqrt{x^2+y^2}}}{\sqrt{x^2+y^2}}dxdy=-\int_0^{2\pi}d\theta\int_1^{\sqrt{2}}e^r dr=-2\pi\left(e^{\sqrt{2}}-e\right)$$

$$\therefore\oiint_\Sigma\frac{e^{\sqrt{2}}}{\sqrt{x^2-y^2}}dxdy=-2\pi e+2\sqrt{2}\pi e^{\sqrt{2}}-2\pi e^{\sqrt{2}}+2\pi e$$

$$=2\pi e^{\sqrt{2}}\left(\sqrt{2}-1\right)$$

18. 证：设$\vec{\alpha}=P\vec{i}+Q\vec{j}+R\vec{k}$，且$P$，$Q$，$R$具有二阶连续导数

$$\because \nabla \propto \vec{\alpha} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\vec{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\vec{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\vec{k}$$

$$\therefore \nabla \propto (\nabla \propto \vec{\alpha}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ \dfrac{\partial P}{\partial y} - \dfrac{\partial Q}{\partial z} & \dfrac{\partial P}{\partial z} - \dfrac{\partial R}{\partial x} & \dfrac{\partial Q}{\partial x} - \dfrac{\partial P}{\partial y} \end{vmatrix}$$

$$= \left(\frac{\partial^2 Q}{\partial x \partial y} - \frac{\partial^2 P}{\partial y^2} - \frac{\partial^2 P}{\partial z^2} + \frac{\partial^2 Q}{\partial x \partial y}\right)\vec{i} + \left(\frac{\partial^2 R}{\partial y \partial z} - \frac{\partial^2 Q}{\partial z^2} - \frac{\partial^2 Q}{\partial x^2} + \frac{\partial^2 P}{\partial y \partial x}\right)\vec{j}$$

$$+ \left(\frac{\partial^2 P}{\partial z \partial x} - \frac{\partial^2 Q}{\partial x^2} - \frac{\partial^2 R}{\partial y^2} + \frac{\partial^2 R}{\partial x \partial y}\right)\vec{k}$$

$$= \left(\frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 Q}{\partial x \partial y} + \frac{\partial^2 R}{\partial x \partial z}\right)\vec{i} + \left(\frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} + \frac{\partial^2 P}{\partial z^2}\right)\vec{i} + \left(\frac{\partial^2 P}{\partial y \partial z} + \frac{\partial^2 Q}{\partial y^2} + \frac{\partial^2 R}{\partial y \partial z}\right)\vec{i}$$

$$- \left(\frac{\partial^2 P}{\partial y \partial z} + \frac{\partial^2 Q}{\partial y^2} + \frac{\partial^2 R}{\partial y \partial z}\right)\vec{j} - \left(\frac{\partial^2 Q}{\partial x^2} + \frac{\partial^2 Q}{\partial y^2} + \frac{\partial^2 Q}{\partial z^2}\right)\vec{j} + \left(\frac{\partial^2 P}{\partial z \partial x} + \frac{\partial^2 Q}{\partial z \partial y} + \frac{\partial^2 R}{\partial z^2}\right)\vec{k}$$

$$- \left(\frac{\partial^2 R}{\partial x^2} + \frac{\partial^2 R}{\partial y^2} + \frac{\partial^2 R}{\partial z^2}\right)\vec{k}$$

$$= \frac{\partial}{\partial x}\left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right)\vec{i} - \nabla^2 P\vec{i} + \frac{\partial}{\partial y}\left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right)\vec{j} - \nabla^2 Q\vec{j}$$

$$+ \frac{\partial}{\partial z}\left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right)\vec{k} - \nabla^2 R\vec{k}$$

$$= \nabla(\nabla\vec{\alpha}) - \nabla^2\vec{\alpha}$$